ASHWANI KUMAR SINGH

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Education

Dr. B. R. Ambedkar National Institute of Technology Jalandhar **Jun 2026**
Bachelor of Technology in Computer Science and Engineering

Experience

CloudFinOps – SDE Intern **Jan 2026 – Jul 2026**

- Built responsive **cloud cost monitoring dashboards** using **React.js, Node.js, and Supabase**, integrating real-time data visualization to track infrastructure spending, resource utilization, and system performance across cloud environments.
- Optimized **PostgreSQL database schemas** through **B-tree indexing, SQL query tuning**, and efficient data retrieval strategies, **reducing query latency by 35%** and improving the responsiveness of analytics dashboards.
- Developed and integrated **RESTful APIs** using **Node.js and Express.js** to manage data exchange between frontend components and backend services, streamline request handling, and support reliable application functionality.
- Automated **CI/CD pipelines** with **GitHub Actions and AWS**, simplifying build, testing, and deployment processes, minimizing manual intervention, and supporting **99.9% application uptime**.

DeepQ-AI – Automation Intern **Jun 2025 – Sept 2025**

- Built an **AI-powered email automation system** using **Python, LangGraph, and LLMs** to classify incoming emails, extract key information, and generate **context-aware response drafts**.
- Integrated **Retrieval-Augmented Generation (RAG)** with **LangChain, FAISS, and semantic search** to retrieve relevant information from a knowledge base and improve response accuracy.
- Designed **multi-step agent workflows** with **LangGraph**, incorporating conditional routing, conversation memory, and **human approval** before sending automated replies through the **Gmail API**.
- Implemented **LangSmith tracing and evaluation** to monitor LLM responses, troubleshoot agent execution, and improve workflow reliability; exposed automation features through **FastAPI endpoints**.

Projects

Workstream | Distributed Task Processing Engine
Technologies: Python, FastAPI, Celery, Redis, Docker, AWS EC2

- Created a **distributed task processing service** using **FastAPI and Celery** to execute background jobs asynchronously, handling **over 520 requests per second**.
- Added **Redis-based rate limiting** with **Lua scripts and dead-letter queues** to manage traffic, handle failed tasks, and improve recovery during processing errors.
- Deployed containerized workers on **AWS EC2** using **Docker** and tuned concurrency settings, increasing processing throughput by **3.8 times**.
- Configured **Celery task queues** with **automatic retries and exponential backoff** to handle transient failures, prevent task loss, and improve the reliability of **asynchronous job execution**.

SearchRank Analytics Engine | Search Evaluation Framework
Technologies: Python, Pandas, NumPy, YAML

- Created a **Python-based search evaluation tool** to measure ranking quality across **100+ queries** using **NDCG@10** and **Mean Average Precision (MAP)**.
- **Reduced data processing time by 45%** by replacing repetitive operations with **vectorized NumPy calculations** and efficient **Pandas transformations**.
- Added **YAML-based configuration**, command-line execution, and **bootstrap confidence intervals** to make experiments repeatable and compare evaluation results.
- Developed **automated unit tests** using **PyTest** to validate ranking metrics, handle edge cases, and ensure consistent results across different search datasets.

Technical Skills

- **Programming Languages:** Java, Python, C++, JavaScript, TypeScript, SQL
- **Backend & Frontend:** Node.js, Express.js, FastAPI, Spring Boot, RESTful APIs, GraphQL, React.js, Next.js, Redux, Tailwind CSS
- **Databases & Cloud:** PostgreSQL, MySQL, MongoDB, Redis, Supabase, AWS EC2, AWS S3, Docker, GitHub Actions, Git, Linux, Nginx
- **CS Fundamentals & Tools:** DSA, OOP, DBMS, OS, CN, Celery, Distributed Systems, CI/CD, PyTest
- **AI & Automation:** LLMs, RAG, LangChain, LangGraph, LangSmith, FAISS, Prompt Engineering, AI Agents

Achievements

- **HackWithInfy:** Participated in a competitive coding challenge, solving algorithmic problems using data structures and problem-solving techniques.
- **College Hackathon:** Collaborated with a team to build and present a working software prototype within a limited timeframe.
- **Competitive Programming:** Solved 600+ coding problems across LeetCode, GeeksforGeeks, and HackerRank, strengthening problem-solving skills in Data Structures, Algorithms, and algorithmic optimization.
- **Machine Learning Pipelines with Azure ML Studio** – Coursera